The Neon Trompo: Voice Ordering Agent

This repository contains the backend infrastructure for a Tex-Mex food truck voice AI, utilizing a Dual-LLM architecture to eliminate latency and hallucination.

Grading Pre-Steps: ngrok & Retell AI

As approved by the professor, to run this system end-to-end with a live telephony webhook, please execute the following manual step to bridge the local Docker container to the public web.

1. **Install ngrok** (brew install ngrok/ngrok/ngrok on macOS, or download from ngrok.com).
2. **Authenticate:** Run ngrok config add-authtoken <your_token> using your free account.
3. **Start the tunnel:** Start the tunnel pointing to the FastAPI port:
   ngrok http 8000

4. **Environment Setup:** * Copy the https://... forwarding URL provided by ngrok.
   - Duplicate .env.example to .env.
   - Paste the URL into RETELL_WEBHOOK_URL in the .env file (if testing Retell live).
   - Ensure a valid GEMINI_API_KEY is present in the .env file.

Execution (Deploy Pillar)

Once the .env is configured, boot the application via Docker:
docker compose up --build

Testing (Stress & Test Pillars)

To run the automated Test suites and the Load test, use the included Makefile commands.
(Note: If running on Windows without make, use pytest tests/user_stories/ -v)
Run User Stories:
make test

Run Load Test (10 req/sec via Locust):
make loadtest

(The load test targets the included Local WebSocket Simulator at ws://localhost:8000/ws/chat to verify the exact Retell AI payload schema without wasting live API credits).